

**Star Milvus
for a chance
to win a prize
tonight!**



This just dropped (open-source edition)!

<https://github.com/canopyai/Orpheus-TTS>

<https://huggingface.co/ds4sd/SmolDocling-256M-preview>

<https://huggingface.co/nvidia/GR00T-N1-2B>

The world is much more than just text and keywords

◀ **90%** of newly generated data in 2025 will be unstructured data ▶



10%
Other

Mission:

Helping organizations make sense of unstructured data.



2017
Founded



\$113M
Raised



140+
Employees



Redwood City, CA
Headquarters

Welcome Speakers

TECH TALK 1

From O1 to O3: How Reasoning Transformed AI and Launched a New Generation of Agents



Harry Zhao
Member of Technical Staff, OpenAI

TECH TALK 2

Building an Agent to Reason about Private Data with OpenAI & Milvus



Stefan Webb
Developer Advocate, Zilliz

Building a Reasoning Agent with Milvus and OpenAI

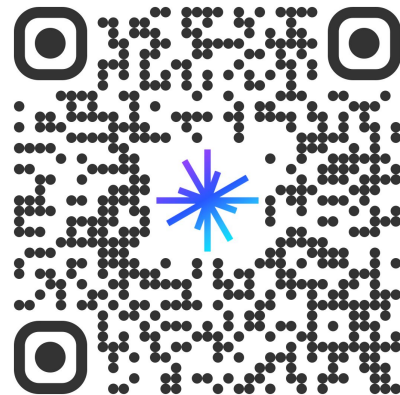


Stefan Webb

Developer Advocate, Zilliz

stefan.webb@zilliz.com

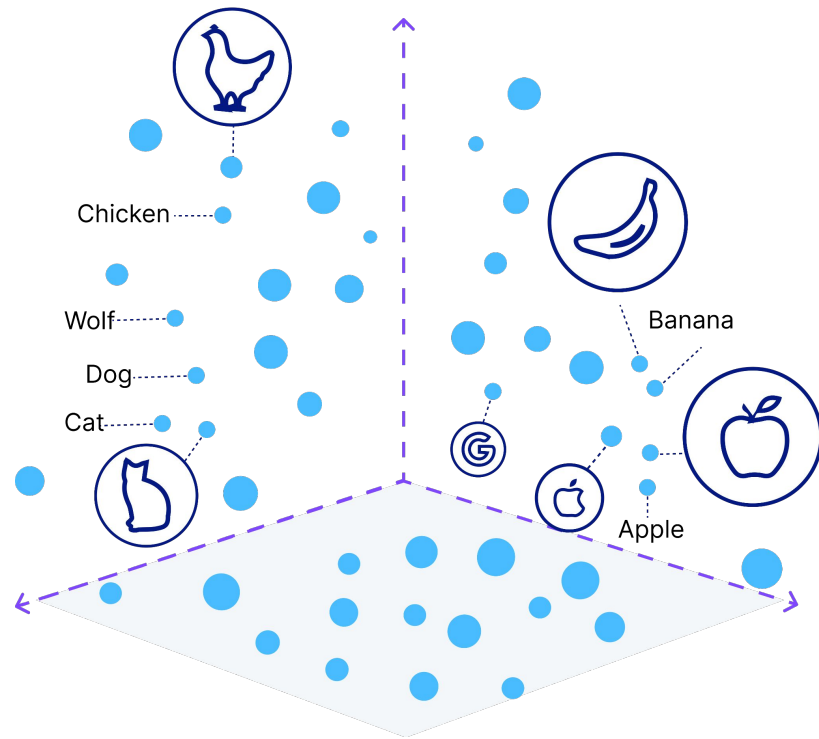
<https://www.linkedin.com/in/stefan-webb>



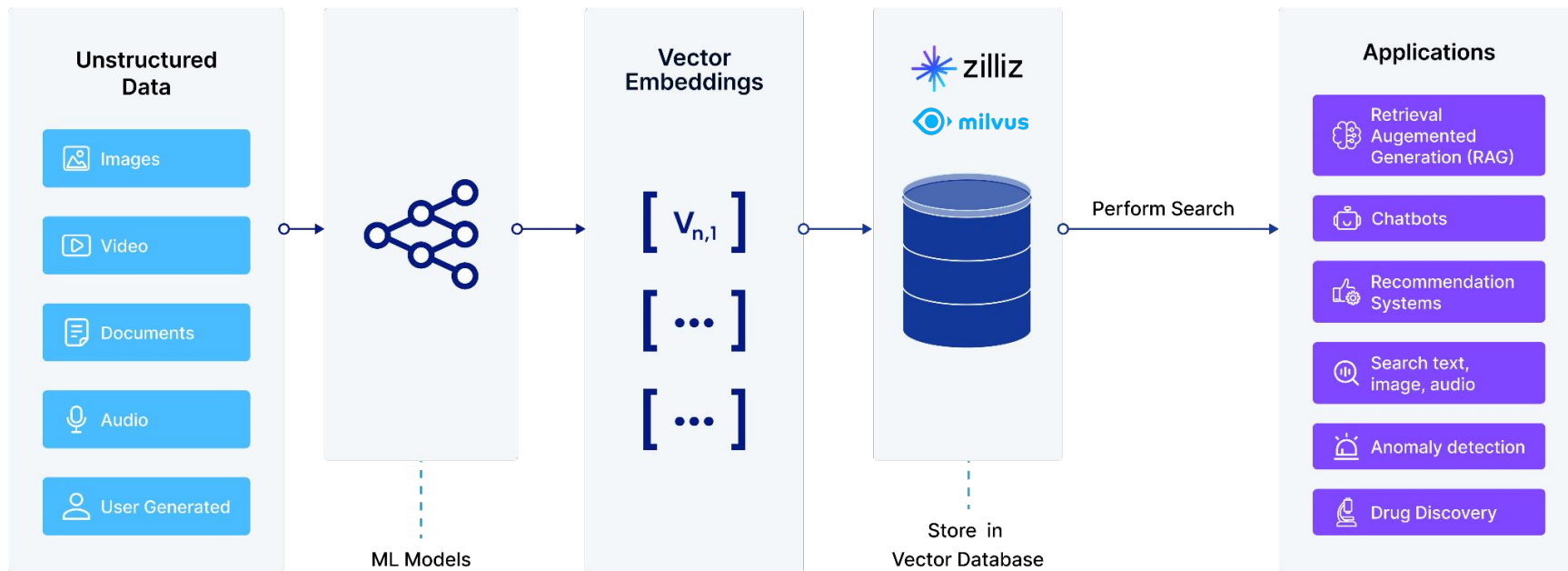
01

Why Milvus?

Vector search is the new standard



A New tool emerged. The Vector Database



Milvus: The most widely-adopted vector database

Milvus is an **Open-Source Vector Database** to **store, index, manage, and use** the massive number of **embedding vectors** generated by deep neural networks and LLMs.



430+

contributors



33K+

stars



67M+

docker pulls



2.8K
+

forks

Built by database & AI experts

Zilliz was built by a top-tier team of **algorithm and database engineers** with a strong pedigree in developing **high-performance, scalable, and highly available** distributed systems, uniquely tailored for **vector search**.

Milvus: A Purpose-Built Vector Data Management System

Jingxiao Wang¹, Xuesong Yu¹, Ruijing Guo¹, Hua Jin¹, Peng Xu¹, Shengxin Li¹, Xuesong Wang¹,
Jingyuan Cai¹, Changping Li¹, Xiaohu Xu¹, Xun Lu¹, Peng Han¹, Yijun Guo¹, Yijun Guo¹,
Yuekang Cai¹, Zhenzhang Li¹, Zhenzhang Zhong¹, Yihua Mo¹, Jia Cai¹, Binliang Li¹, Li Wang¹, Charles Sun¹
¹Zilliz & Alibaba

²OpenMMLab

³OpenMMLab

⁴OpenMMLab

⁵OpenMMLab

⁶OpenMMLab

⁷OpenMMLab

⁸OpenMMLab

⁹OpenMMLab

¹⁰OpenMMLab

¹¹OpenMMLab

¹²OpenMMLab

¹³OpenMMLab

¹⁴OpenMMLab

¹⁵OpenMMLab

¹⁶OpenMMLab

¹⁷OpenMMLab

¹⁸OpenMMLab

¹⁹OpenMMLab

²⁰OpenMMLab

²¹OpenMMLab

²²OpenMMLab

²³OpenMMLab

²⁴OpenMMLab

²⁵OpenMMLab

²⁶OpenMMLab

²⁷OpenMMLab

²⁸OpenMMLab

²⁹OpenMMLab

³⁰OpenMMLab

³¹OpenMMLab

³²OpenMMLab

³³OpenMMLab

³⁴OpenMMLab

³⁵OpenMMLab

³⁶OpenMMLab

³⁷OpenMMLab

³⁸OpenMMLab

³⁹OpenMMLab

⁴⁰OpenMMLab

⁴¹OpenMMLab

⁴²OpenMMLab

⁴³OpenMMLab

⁴⁴OpenMMLab

⁴⁵OpenMMLab

⁴⁶OpenMMLab

⁴⁷OpenMMLab

⁴⁸OpenMMLab

⁴⁹OpenMMLab

⁵⁰OpenMMLab

⁵¹OpenMMLab

⁵²OpenMMLab

⁵³OpenMMLab

⁵⁴OpenMMLab

⁵⁵OpenMMLab

⁵⁶OpenMMLab

⁵⁷OpenMMLab

⁵⁸OpenMMLab

⁵⁹OpenMMLab

⁶⁰OpenMMLab

⁶¹OpenMMLab

⁶²OpenMMLab

⁶³OpenMMLab

⁶⁴OpenMMLab

⁶⁵OpenMMLab

⁶⁶OpenMMLab

⁶⁷OpenMMLab

⁶⁸OpenMMLab

⁶⁹OpenMMLab

⁷⁰OpenMMLab

⁷¹OpenMMLab

⁷²OpenMMLab

⁷³OpenMMLab

⁷⁴OpenMMLab

⁷⁵OpenMMLab

⁷⁶OpenMMLab

⁷⁷OpenMMLab

⁷⁸OpenMMLab

⁷⁹OpenMMLab

⁸⁰OpenMMLab

⁸¹OpenMMLab

⁸²OpenMMLab

⁸³OpenMMLab

⁸⁴OpenMMLab

⁸⁵OpenMMLab

⁸⁶OpenMMLab

⁸⁷OpenMMLab

⁸⁸OpenMMLab

⁸⁹OpenMMLab

⁹⁰OpenMMLab

⁹¹OpenMMLab

⁹²OpenMMLab

⁹³OpenMMLab

⁹⁴OpenMMLab

⁹⁵OpenMMLab

⁹⁶OpenMMLab

⁹⁷OpenMMLab

⁹⁸OpenMMLab

⁹⁹OpenMMLab

¹⁰⁰OpenMMLab

¹⁰¹OpenMMLab

¹⁰²OpenMMLab

¹⁰³OpenMMLab

¹⁰⁴OpenMMLab

¹⁰⁵OpenMMLab

¹⁰⁶OpenMMLab

¹⁰⁷OpenMMLab

¹⁰⁸OpenMMLab

¹⁰⁹OpenMMLab

¹¹⁰OpenMMLab

¹¹¹OpenMMLab

¹¹²OpenMMLab

¹¹³OpenMMLab

¹¹⁴OpenMMLab

¹¹⁵OpenMMLab

¹¹⁶OpenMMLab

¹¹⁷OpenMMLab

¹¹⁸OpenMMLab

¹¹⁹OpenMMLab

¹²⁰OpenMMLab

¹²¹OpenMMLab

¹²²OpenMMLab

¹²³OpenMMLab

¹²⁴OpenMMLab

¹²⁵OpenMMLab

¹²⁶OpenMMLab

¹²⁷OpenMMLab

¹²⁸OpenMMLab

¹²⁹OpenMMLab

¹³⁰OpenMMLab

¹³¹OpenMMLab

¹³²OpenMMLab

¹³³OpenMMLab

¹³⁴OpenMMLab

¹³⁵OpenMMLab

¹³⁶OpenMMLab

¹³⁷OpenMMLab

¹³⁸OpenMMLab

¹³⁹OpenMMLab

¹⁴⁰OpenMMLab

¹⁴¹OpenMMLab

¹⁴²OpenMMLab

¹⁴³OpenMMLab

¹⁴⁴OpenMMLab

¹⁴⁵OpenMMLab

¹⁴⁶OpenMMLab

¹⁴⁷OpenMMLab

¹⁴⁸OpenMMLab

¹⁴⁹OpenMMLab

¹⁵⁰OpenMMLab

¹⁵¹OpenMMLab

¹⁵²OpenMMLab

¹⁵³OpenMMLab

¹⁵⁴OpenMMLab

¹⁵⁵OpenMMLab

¹⁵⁶OpenMMLab

¹⁵⁷OpenMMLab

¹⁵⁸OpenMMLab

¹⁵⁹OpenMMLab

¹⁶⁰OpenMMLab

¹⁶¹OpenMMLab

¹⁶²OpenMMLab

¹⁶³OpenMMLab

¹⁶⁴OpenMMLab

¹⁶⁵OpenMMLab

¹⁶⁶OpenMMLab

¹⁶⁷OpenMMLab

¹⁶⁸OpenMMLab

¹⁶⁹OpenMMLab

¹⁷⁰OpenMMLab

¹⁷¹OpenMMLab

¹⁷²OpenMMLab

¹⁷³OpenMMLab

¹⁷⁴OpenMMLab

¹⁷⁵OpenMMLab

¹⁷⁶OpenMMLab

¹⁷⁷OpenMMLab

¹⁷⁸OpenMMLab

¹⁷⁹OpenMMLab

¹⁸⁰OpenMMLab

¹⁸¹OpenMMLab

¹⁸²OpenMMLab

¹⁸³OpenMMLab

¹⁸⁴OpenMMLab

¹⁸⁵OpenMMLab

¹⁸⁶OpenMMLab

¹⁸⁷OpenMMLab

¹⁸⁸OpenMMLab

¹⁸⁹OpenMMLab

¹⁹⁰OpenMMLab

¹⁹¹OpenMMLab

¹⁹²OpenMMLab

¹⁹³OpenMMLab

¹⁹⁴OpenMMLab

¹⁹⁵OpenMMLab

¹⁹⁶OpenMMLab

¹⁹⁷OpenMMLab

¹⁹⁸OpenMMLab

¹⁹⁹OpenMMLab

²⁰⁰OpenMMLab

²⁰¹OpenMMLab

²⁰²OpenMMLab

²⁰³OpenMMLab

²⁰⁴OpenMMLab

²⁰⁵OpenMMLab

²⁰⁶OpenMMLab

²⁰⁷OpenMMLab

²⁰⁸OpenMMLab

²⁰⁹OpenMMLab

²¹⁰OpenMMLab

²¹¹OpenMMLab

²¹²OpenMMLab

²¹³OpenMMLab

²¹⁴OpenMMLab

²¹⁵OpenMMLab

²¹⁶OpenMMLab

²¹⁷OpenMMLab

²¹⁸OpenMMLab

²¹⁹OpenMMLab

²²⁰OpenMMLab

²²¹OpenMMLab

²²²OpenMMLab

²²³OpenMMLab

²²⁴OpenMMLab

²²⁵OpenMMLab

²²⁶OpenMMLab

²²⁷OpenMMLab

²²⁸OpenMMLab

²²⁹OpenMMLab

²³⁰OpenMMLab

²³¹OpenMMLab

²³²OpenMMLab

²³³OpenMMLab

²³⁴OpenMMLab

²³⁵OpenMMLab

²³⁶OpenMMLab

²³⁷OpenMMLab

²³⁸OpenMMLab

²³⁹OpenMMLab

²⁴⁰OpenMMLab

²⁴¹OpenMMLab

²⁴²OpenMMLab

²⁴³OpenMMLab

²⁴⁴OpenMMLab

²⁴⁵OpenMMLab

²⁴⁶OpenMMLab

²⁴⁷OpenMMLab

²⁴⁸OpenMMLab

²⁴⁹OpenMMLab

²⁵⁰OpenMMLab

²⁵¹OpenMMLab

²⁵²OpenMMLab

²⁵³OpenMMLab

²⁵⁴OpenMMLab

²⁵⁵OpenMMLab

²⁵⁶OpenMMLab

²⁵⁷OpenMMLab

²⁵⁸OpenMMLab

²⁵⁹OpenMMLab

²⁶⁰OpenMMLab

²⁶¹OpenMMLab

²⁶²OpenMMLab

²⁶³OpenMMLab

²⁶⁴OpenMMLab

²⁶⁵OpenMMLab

²⁶⁶OpenMMLab

²⁶⁷OpenMMLab

²⁶⁸OpenMMLab

²⁶⁹OpenMMLab

²⁷⁰OpenMMLab

²⁷¹OpenMMLab

²⁷²OpenMMLab

²⁷³OpenMMLab

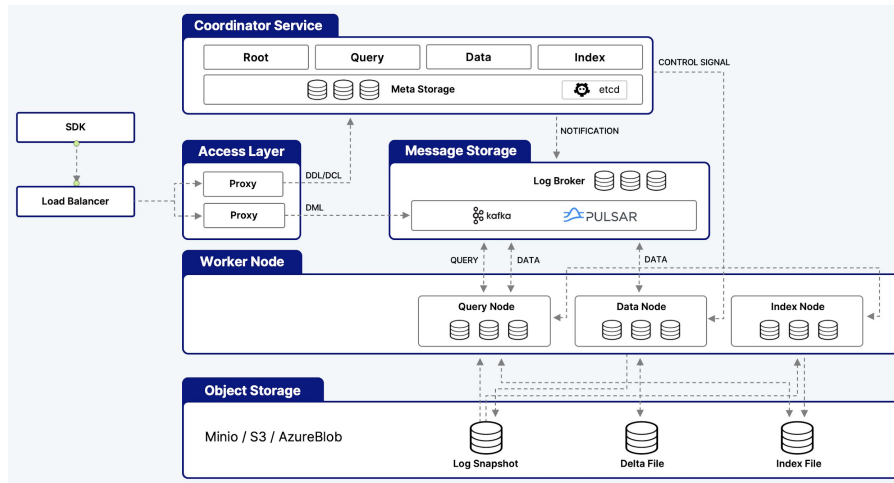
²⁷⁴OpenMMLab

Milvus Architecture

Design Principles

- Separation of storage and compute
- Fully depend on mature storage systems
- Microservice - scale by functionality
- Separate streaming and historical data
- Pluggable engine, storage and index
- Log as data

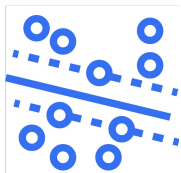
Fully distributed, designed for scalability



Rich functionality



Dynamic Schema



Float, Binary, &
Sparse Vector



Tag+Vector
Optimized Filtering



Hybrid Search
Dense & Sparse



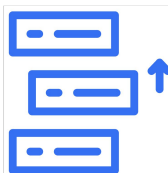
RBAC, TLS,
Encryption



Million+ level
tenant support



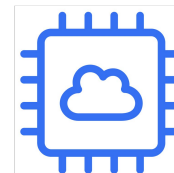
Disk Based
Index



Tiered Storage



Bulk Import



GPU, Intel & ARM
CPU support

Industry leaders already use vector search in their apps

Use Case: Data Search

Vectors: 2 Billion

Req'ts: 200 ms, Cost mgmt

Index: DiskANN for cost savings

Use Case: Drug Discovery

Vectors: 12 Billion

Req'ts: High Recall

Index: BIN_FLAT

Use Case: Image Search

Vectors: 20 Billion

Req'ts: High Insertion, Cost

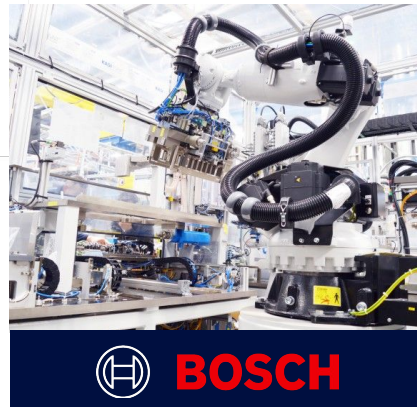
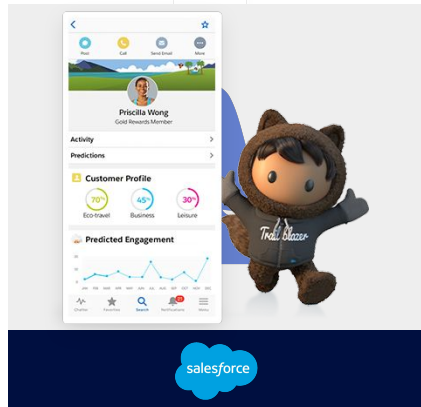
Index: Disk Based Index

Use Case: Recommender System

Vectors: 20 Billion

Req'ts: 5,000 QPS

Index: HNSW & CAGRA



Milvus Users

accenture

airbnb

AT&T



BOSCH

Chegg

CISCO

CISION

COMPASS

Deloitte.

ebay

FARFETCH

Grab

IKEA

Inflection

intuit

Microsoft

new relic

nVIDIA

OMERS

Otter.ai

PayPal

paloalto
NETWORKS

POSHMARK

ROBLOX

salesforce

Shell

shutterstock

T

TREND
MICRO

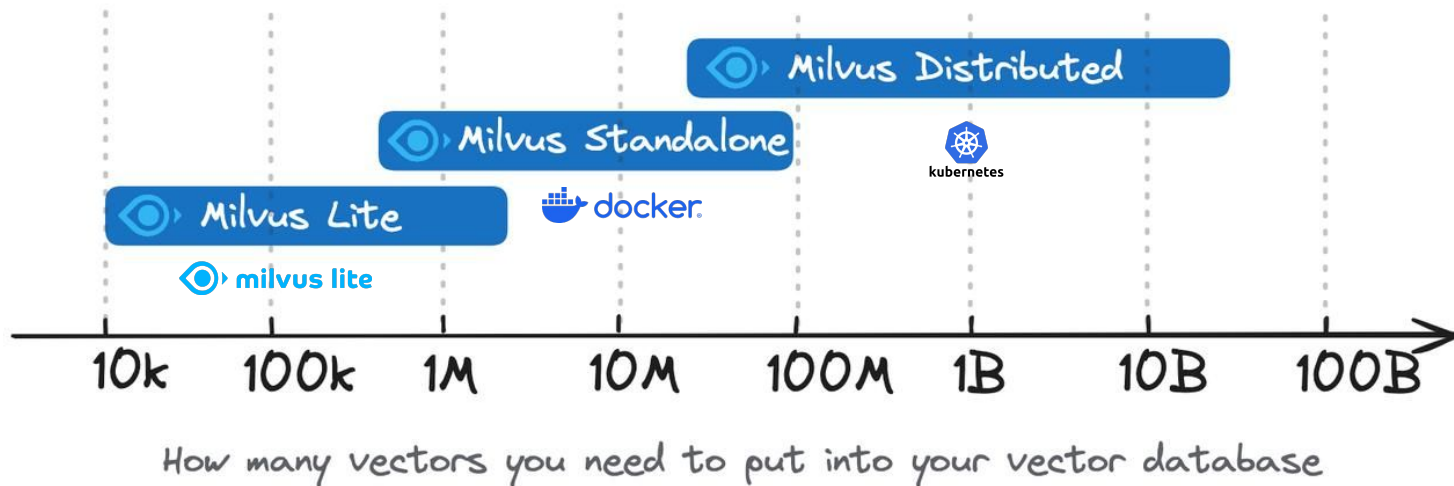
Walmart



ZipRecruiter

zomato

Flexible Deployment Options for Milvus



Set up Once: Common API across all products regardless of architecture

AI Stack



Software Infrastructure

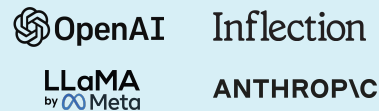
Embedding Models



Vector Database



LLMs



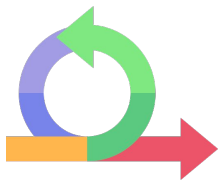
Hardware Infrastructure



02

DeepSearcher demo

Research Agents

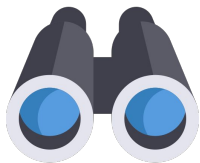


Iteration

“...learned to plan and execute a multi-step trajectory...”

“...backtracking and reacting to real-time information...”

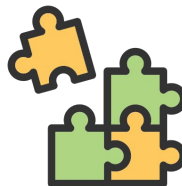
“...pivoting as needed in reaction to information it encounters...”



Search

“...trained using end-to-end reinforcement learning on hard browsing and reasoning tasks across a range of domains...”

“...optimized for web browsing and data analysis...”

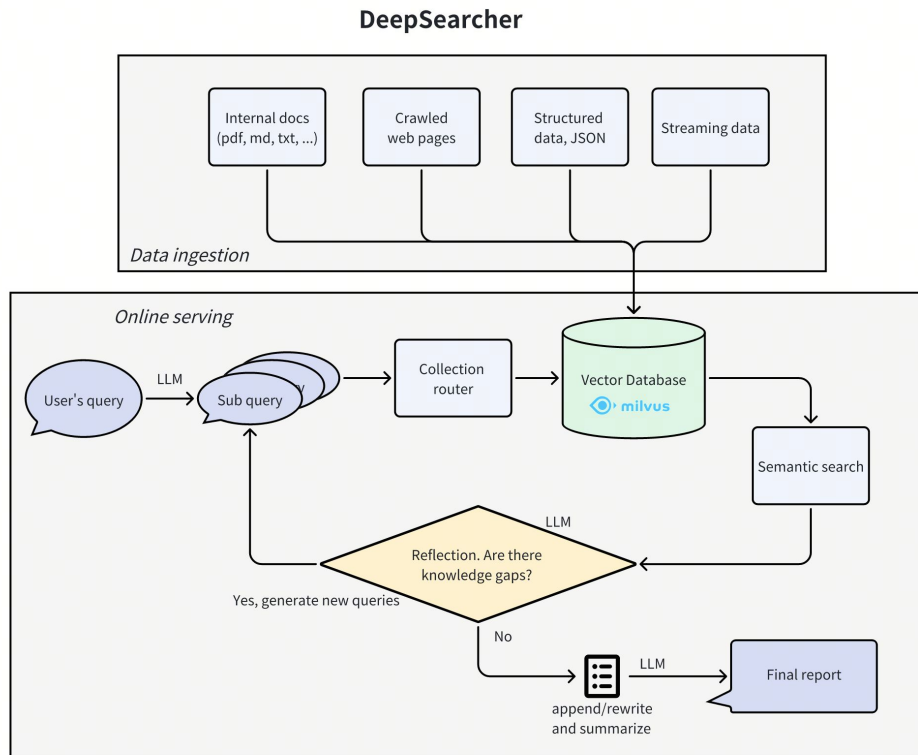


Reasoning

“...fine-tuned on the upcoming OpenAI o3 reasoning model...”

““...leverages reasoning to search, interpret, and analyze massive amounts of text...””

DeepSearcher from Zilliz




```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent % deepsearcher --query "Write a report comparing Milvus with other vector databases"
```

```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher --query "Write a report comparing Milvus with other vector databases"
<query> Write a report comparing Milvus with other vector databases </query>
```

our query/task

```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher —query "Write a report comparing Milvus with other vector databases"
```

```
<query> Write a report comparing Milvus with other vector databases </query>
```

```
<think> Break down the original query into new sub queries: ['What is Milvus?', 'What are the key features of Milvus?', 'What are some other popular vector databases?', 'How does Milvus compare with these other vector databases in terms of performance?', 'What are the use cases for Milvus and other vector databases?', 'What are the advantages and disadvantages of using Milvus compared to other vector databases?', 'How is the community support and documentation for Milvus versus other vector databases?', 'What are the scalability and deployment options for Milvus compared to other vector databases?']</think>
```

```
>> Iteration: 1
```

```
|
```

initial subqueries

```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher —query "Write a report comparing Milvus with other vector databases"
<query> Write a report comparing Milvus with other vector databases </query>

<think> Break down the original query into new sub queries: ['What is Milvus?', 'What are the key features of Milvus?', 'What are some other popular vector databases?', 'How does Milvus compare with these other vector databases in terms of performance?', 'What are the use cases for Milvus and other vector databases?', 'What are the advantages and disadvantages of using Milvus compared to other vector databases?', 'How is the community support and documentation for Milvus versus other vector databases?', 'What are the scalability and deployment options for Milvus compared to other vector databases?']</think>

>> Iteration: 1

<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think>
>

<search> Search [What is Milvus?] in [milvus_docs]... </search>
```

**think what action to perform
search 1st subquery, 1st source**

```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher —query "Write a report comparing Milvus with other vector databases"
<query> Write a report comparing Milvus with other vector databases </query>

<think> Break down the original query into new sub queries: ['What is Milvus?', 'What are the key features of Milvus?', 'What are some other popular vector databases?', 'How does Milvus compare with these other vector databases in terms of performance?', 'What are the use cases for Milvus and other vector databases?', 'What are the advantages and disadvantages of using Milvus compared to other vector databases?', 'How is the community support and documentation for Milvus versus other vector databases?', 'What are the scalability and deployment options for Milvus compared to other vector databases?']</think>

>> Iteration: 1

<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think>
>

<search> Search [What is Milvus?] in [milvus_docs]... </search>

<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/roadmap.md'] </search>

<search> Search [What is Milvus?] in [what_is_milvus]... </search>
```

retrieve chunks with Milvus


```
(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %deepsearcher —query "Write a report comparing Milvus with
other vector databases"
<query> Write a report comparing Milvus with other vector databases </query>

<think> Break down the original query into new sub queries: ['What is Milvus?', 'What are the key features of Mi
lvus?', 'What are some other popular vector databases?', 'How does Milvus compare with these other vector databa
ses in terms of performance?', 'What are the use cases for Milvus and other vector databases?', 'What are the ad
vantages and disadvantages of using Milvus compared to other vector databases?', 'How is the community support a
nd documentation for Milvus versus other vector databases?', 'What are the scalability and deployment options fo
r Milvus compared to other vector databases?']</think>

>> Iteration: 1

<think> Perform search [What is Milvus?] on the vector DB collection: ['milvus-docs', 'what_is_milvus'] </think>
>

<search> Search [What is Milvus?] in milvus-docs... </search>

<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/o
verview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md', '/Users/zilli
z/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.
x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/roadmap.md'] </search>

<search> Search [What is Milvus?] in [what_is_milvus]... </search>

<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatisMilvus.pdf', './examples/data/Whati
sMilvus.pdf', './examples/data/WhatisMilvus.pdf'] </search>
```

search 1st subquery, 2nd source
retrieve chunks with Milvus

```
<think> Break down the original query into new sub queries: ['What is Milvus?', 'What are the key features of Milvus?', 'What are some other popular vector databases?', 'How does Milvus compare with these other vector databases in terms of performance?', 'What are the use cases for Milvus and other vector databases?', 'What are the advantages and disadvantages of using Milvus compared to other vector databases?', 'How is the community support and documentation for Milvus versus other vector databases?', 'What are the scalability and deployment options for Milvus compared to other vector databases?']</think>
```

```
>> Iteration: 1
```

```
<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What is Milvus?] in [milvus_docs]... </search>
```

```
<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/roadmap.md'] </search>
```

```
<search> Search [What is Milvus?] in [what_is_milvus]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>
```

```
<think> Perform search [What are the key features of Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What are the key features of Milvus?] in [milvus_docs]... </search>
```

think what action to perform
search 2nd subquery, 1st source

```
>> Iteration: 1
```

```
<think> Perform search [What is Milvus?] on the vector DB collections: ['milvus_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What is Milvus?] in [milvus_docs]... </search>
```

```
<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/roadmap.md'] </search>
```

```
<search> Search [What is Milvus?] in [what_is_milvus]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>
```

```
<think> Perform search [What are the key features of Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

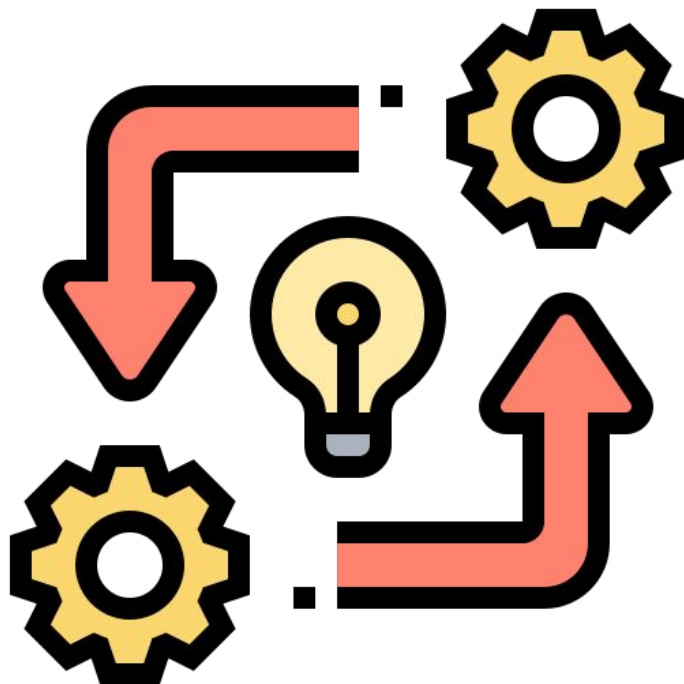
```
<search> Search [What are the key features of Milvus?] in [milvus_docs]... </search>
```

```
<search> Accept 4 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/architecture/architecture.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md'] </search>
```

```
<search> Search [What are the key features of Milvus?] in [milvus_embedding_docs]... </search>
```

Retrieve chunks with Milvus





```
<search> Search [What are the scalability and deployment options for Milvus compared to other vector databases?] in [what_is_milvus]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>
```

```
<search> Search [What are the scalability and deployment options for Milvus compared to other vector databases?] in [milvus_docs]... </search>
```

```
<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/architecture/architecture_overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md'] </search>
```

```
<think> Reflecting on the search results... </think>
```

```
<think> New search queries for next iteration: ['What are the performance benchmarks of Milvus compared to its competitors?', 'What specific use cases are best suited for other vector databases compared to Milvus?', 'What are the recent developments and future directions in vector databases beyond Milvus?'] </think>
```

```
>> Iteration: 2
```

```
<think> Perform search [What are the performance benchmarks of Milvus compared to its competitors?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_docs]... </search>
```

Accelerated Playback

reflect on iter 1, new subqueries

```
<search> Search [What are the scalability and deployment options for Milvus compared to other vector databases?] in [milvus_docs]... </search>
```

```
<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/accelerated-playback.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/architecture/architecture_overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md'] </search>
```

```
<think> Reflecting on the search results... </think>
```

```
<think> New search queries for next iteration: ['What are the performance benchmarks of Milvus compared to its competitors?', 'What specific use cases are best suited for other vector databases compared to Milvus?', 'What are the recent developments and future directions in vector databases beyond Milvus?'] </think>
```

```
>> Iteration: 2
```

```
<think> Perform search [What are the performance benchmarks of Milvus compared to its competitors?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_docs]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/benchmark.md'] </search>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_embedding_docs]... </search>
```

Accelerated Playback

search 1st subquery




```
<search> Accept 5 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/integrations/apify_milvus_rag.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/architecture/architecture_overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md'] </search>
```

Accelerated Playback

```
<think> Reflecting on the search results... </think>
```

```
<think> New search queries for next iteration: ['What are the performance benchmarks of Milvus compared to its competitors?', 'What specific use cases are best suited for other vector databases compared to Milvus?', 'What are the recent developments and future directions in vector databases beyond Milvus?'] </think>
```

```
>> Iteration: 2
```

```
<think> Perform search [What are the performance benchmarks of Milvus compared to its competitors?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_docs]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/benchmark.md'] </search>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_embedding_docs]... </search>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [what_is_milvus]... </search>
```

```
x/site/en/reference/architecture/architecture_overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/getstarted/install-overview.md'] </search>
```

```
<think> Reflecting on the search results... </think>
```

Accelerated Playback

```
<think> New search queries for next iteration: ['What are the performance benchmarks of Milvus compared to its competitors?', 'What specific use cases are best suited for other vector databases compared to Milvus?', 'What are the recent developments and future directions in vector databases beyond Milvus?'] </think>
```

```
>> Iteration: 2
```

```
<think> Perform search [What are the performance benchmarks of Milvus compared to its competitors?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_docs]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/benchmark.md'] </search>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_embedding_docs]... </search>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [what_is_milvus]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>
```

```
ompetitors?', 'What specific use cases are best suited for other vector databases compared to Milvus?', 'What are the recent developments and future directions in vector databases beyond Milvus?'] </think>
```

```
>> Iteration: 2
```

```
<think> Perform search [What are the performance benchmarks of Milvus compared to its competitors?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

Accelerated Playback

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_docs]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/benchmark.md'] </search>
```

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_embedding_docs]... </search>
```

search 2nd subquery

```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [what_is_milvus]... </search>
```

```
<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>
```

```
<think> Perform search [What specific use cases are best suited for other vector databases compared to Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_docs]... </search>
```



```
r DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_docs]...
</search>

<search> Accept 3 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/c
omparison.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/m
ilvus-docs-2.5.x/site/en/reference/benchmark.md'] </search>

<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_embeddin
g_docs]... </search>

<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [what_is_milvus]
... </search>

<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/Whati
sMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>

<think> Perform search [What specific use cases are best suited for other vector databases compared to Milvus?]
on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [mil
vus_docs]... </search>

<search> Accept 1 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/c
omparison.md'] </search>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [mil
vus_embedding_docs]... </search>
```

Accelerated Playback

</search>

<search> Accept 3 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/overview.md', '/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/reference/benchmark.md'] </search>

Accelerated Playback

<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_embedding_docs]... </search>

<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [what_is_milvus]... </search>

<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>

<think> Perform search [What specific use cases are best suited for other vector databases compared to Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_docs]... </search>

<search> Accept 1 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md'] </search>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_embedding_docs]... </search>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [what_is_milvus]... </search>


```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [milvus_embedding_docs]... </search>

<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [what_is_milvus]... </search>

<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>

<think> Perform search [What specific use cases are best suited for other vector databases compared to Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_docs]... </search>

<search> Accept 1 document chunk(s) from references: ['./Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md'] </search>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_embedding_docs]... </search>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [what_is_milvus]... </search>

<think> Perform search [What are the recent developments and future directions in vector databases beyond Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [milvus_docs]... </search>
```

Accelerated Playback

search 3rd subquery



```
<search> Search [What are the performance benchmarks of Milvus compared to its competitors?] in [what_is_milvus] ... </search>

<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>

<think> Perform search [What specific use cases are best suited for other vector databases compared to Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_docs]... </search>

<search> Accept 1 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/compare.md'] </search>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_embedding_docs]... </search>

<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [what_is_milvus]... </search>

<think> Perform search [What are the recent developments and future directions in vector databases beyond Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>

<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [milvus_docs]... </search>

<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [milvus_embedding_docs]... </search>
```

Accelerated Playback

```
<search> Accept 3 document chunk(s) from references: ['./examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf', './examples/data/WhatIsMilvus.pdf'] </search>
```

```
<think> Perform search [What specific use cases are best suited for other vector databases compared to Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

Accelerated Playback

```
<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_docs]... </search>
```

```
<search> Accept 1 document chunk(s) from references: ['/Users/zilliz/Downloads/milvus-docs-2.5.x/site/en/about/comparison.md'] </search>
```

```
<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_embedding_docs]... </search>
```

```
<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [what_is_milvus]... </search>
```

```
<think> Perform search [What are the recent developments and future directions in vector databases beyond Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [milvus_docs]... </search>
```

```
<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [milvus_embedding_docs]... </search>
```

```
<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [what_is_milvus]... </search>
```



```
omparison.md'] </search>
```

```
<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [milvus_embedding_docs]... </search>
```

Accelerated Playback

```
<search> Search [What specific use cases are best suited for other vector databases compared to Milvus?] in [what_is_milvus]... </search>
```

```
<think> Perform search [What are the recent developments and future directions in vector databases beyond Milvus?] on the vector DB collections: ['milvus_docs', 'milvus_embedding_docs', 'what_is_milvus'] </think>
```

```
<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [milvus_docs]... </search>
```

```
<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [milvus_embedding_docs]... </search>
```

```
<search> Search [What are the recent developments and future directions in vector databases beyond Milvus?] in [what_is_milvus]... </search>
```

```
<search> Accept 1 document chunk($1) from references: ['./examples/data/WhatIsMilvus.pdf'] </search>
```

```
<think> Reflecting on the search results... </think>
```

```
<think> No new search queries were generated. Exiting. </think>
```

```
<think> Generating final answer... </think>
```

```
<think> Summarize answer from all 21 retrieved chunks... </think>
```

reflect on iter 2,
no new subqueries

```
<think> No new search queries were generated. Exiting. </think>
```

```
<think> Generating final answer... </think>
```

```
<think> Summarize answer from all 21 retrieved chunks... </think>
```

← generate report

```
=== FINAL ANSWER===
```

```
### Report: Comparison of Milvus with Other Vector Databases
```

```
**Introduction to Milvus:**
```

Milvus is a high-performance, scalable vector database designed for managing unstructured data like images, text, and audio by converting them into numerical vectors. It is open-source and available as a cloud service, supporting various deployment options: Milvus Lite, Standalone, and Distributed. These variants cater to different scales, from local testing environments to large Kubernetes deployments capable of handling tens of billions of vectors.

```
**Key Features of Milvus:**
```

1. ****High Performance****: Milvus is optimized for speed and efficiency. It employs advanced data access patterns and hardware-aware optimizations, achieving a performance boost of 30%-70% over popular implementations like FAISS and HNSWLib.
2. ****Flexible Deployment****: It supports various deployment options—Milvus Lite for lightweight applications, Standalone for single-host setups, and Distributed architecture for high-scale scenarios.
3. ****Robust Indexing Algorithms****: Milvus supports diverse indexing/search algorithms (IVF, HNSW, DiskANN) enhancing query speed through real-time processing.
4. ****Advanced Querying Capabilities****: The database supports multiple types of searches, including ANN, hybrid, and filtered searches, alongside maintaining support for advanced data types like sparse and binary vectors.

```
**Comparative Analysis:**
```

2. ****Flexible Deployment****: It supports various deployment options—Milvus Lite for lightweight applications, Standalone for single-host setups, and Distributed architecture for high-scale scenarios.
3. ****Robust Indexing Algorithms****: Milvus supports diverse indexing/search algorithms (IVF, HNSW, DiskANN) enhancing query speed through real-time processing.
4. ****Advanced Querying Capabilities****: The database supports multiple types of searches, including ANN, hybrid, and filtered searches, alongside maintaining support for advanced data types like sparse and binary vectors.

****Comparative Analysis****

1. ****Performance****: Milvus often outperforms its competitors (like Pinecone and others) by a factor of 2-5x due to its efficient engine implemented in C++, optimized for low-level resource management and multi-threading.
2. ****Scalability****: With its cloud-native architecture, Milvus scales linearly to handle significant loads and supports concurrent queries efficiently. It managed billion-scale vectors with minimal performance degradation during extensive testing.
3. ****Community Support and Documentation****: Milvus has robust community engagement and extensive documentation compared to many other vector databases. Its integrations and user contributions, especially from large corporations, enhance its reliability and development support.
4. ****Use Cases****: Milvus excels in applications needing intense similarity searches, such as image retrieval, AI-driven applications, and machine learning frameworks. However, some scenarios may better suit specialized databases like Pinecone for specific niche applications.

****Advantages and Disadvantages****

- **Advantages**

- Superior performance and speed.
- Flexible regarding deployment options.
- Comprehensive SDK support and community-driven documentation.

- **Disadvantages**

- Complexity in setup for new users.
- Requires careful optimization for specific workloads.

****Recent Developments and Future Directions****

Milvus is continuously evolving with planned updates that focus on enhancing its speed, adopting more advanced

****Comparative Analysis:****

1. ****Performance****: Milvus often outperforms its competitors (like Pinecone and others) by a factor of 2-5x due to its efficient engine implemented in C++, optimized for low-level resource management and multi-threading.
2. ****Scalability****: With its cloud-native architecture, Milvus scales linearly to handle significant loads and supports concurrent queries efficiently. It managed billion-scale vectors with minimal performance degradation during extensive testing.
3. ****Community Support and Documentation****: Milvus has robust community engagement and extensive documentation compared to many other vector databases. Its integrations and user contributions, especially from large corporations, enhance its reliability and development support.
4. ****Use Cases****: Milvus excels in applications needing intense similarity searches, such as image retrieval, AI-driven applications, and machine learning frameworks. However, some scenarios may better suit specialized databases like Pinecone for specific niche applications.

****Advantages and Disadvantages:****

- ****Advantages****:
 - Superior performance and speed.
 - Flexible regarding deployment options.
 - Comprehensive SDK support and community-driven documentation.
- ****Disadvantages****:
 - Complexity in setup for new users.
 - Requires careful optimization for specific workloads.

****Recent Developments and Future Directions:****

Milvus is continuously evolving with planned updates that focus on enhancing its speed, adopting more advanced indexing mechanisms, and exploring serverless architectures for even greater scalability and cost-effectiveness.

****Conclusion:****

Milvus is a compelling choice among vector databases, noteworthy for its high performance, flexible deployment options, and strong community support. While it shines in many applications, specific use cases might warrant looking into tailored alternatives depending on organizational needs and operational scale.

(py310) zilliz@zillizdeMacBook-Pro-6 deep-rag-agent %

03

Let's build it ourselves!



It's Coding Time!

<https://bit.ly/41HjZbJ>



04

Where to from here?

Where to from here?



Data

- Open-source
- High-quality
- Reasoning traces



Cost

- Specialized hardware
- Algorithmic improvements



Scaling

- Per-se

Zilliz Offerings

<https://cloud.zilliz.com/signup>

SELF MANAGED SOFTWARE



Milvus

Most widely-adopted open source vector database



FULLY MANAGED SERVICE



Zilliz Cloud

AI Powered Search that is performant and scales



Google Cloud



Azure

BRING YOUR OWN CLOUD



Zilliz BYOC

For Private VPCs



Google Cloud



Azure
Coming Soon!



Set up Once: Common API across all products regardless of architecture

LET'S STAY CONNECTED!

Stefan Webb

Developer Advocate, Zilliz



<https://milvus.io/discord>

Book a Free 1:1 Session

For Support Productionizing Milvus



<http://bit.ly/43YkEYW>